De La Salle University

Modeling Degree ROI Through University Financial Data

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Data Mining

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1 Introduction

The cost of higher education has grown faster than household incomes for decades, turning the decision to enroll into a financial calculation as much as an academic one. Students and families weigh tuition against an expected payoff, yet the payoff is hard to observe at the moment of choice. Published tuition is the most visible price signal available, and it is natural to ask whether institutions that charge more also produce graduates who earn more. This study tests that question directly using institution-level data from the United States Department of Education College Scorecard, which reports tuition alongside the median earnings of former students ten years after entry.

The relationship between tuition and earnings is not self-evidently uniform. Public, private nonprofit, and for-profit institutions operate on different cost structures and serve different student populations, so the same tuition increase may carry very different earnings implications across sectors. The problem this study addresses is whether tuition is a reliable signal of post-graduation earnings at all, and whether that signal, if it exists, is consistent across institutional types. Establishing this empirically matters for students comparing institutions and for policymakers evaluating transparency and accountability in higher education pricing.

The scope is deliberately institutional. The analysis uses a cleaned subset of the College Scorecard covering 2,894 institutions with valid tuition, earnings, and debt records, drawn from an initial sample of 6,273. The central variables are in-state tuition and median earnings ten years after entry, with institutional control type, graduate debt, and undergraduate enrollment providing context. The study proceeds in three stages: descriptive and exploratory analysis of sector-level patterns, a rank-based test of association, and a sequence of regression models that estimate the tuition-earnings relationship overall and by sector, with robustness checks for institutional characteristics. The objective is a descriptive answer to the research question, not a causal estimate of the returns to tuition.

1.1 Objectives

The overarching objective of this investigation is to evaluate the economic returns of postsecondary education by analyzing the relationship between institutional pricing and long-term labor-market outcomes. Postsecondary enrollment is increasingly evaluated through an investment framework in which prospective students and families weigh published tuition costs against expected graduate earnings. Classical human capital models posit that higher educational expenditures reflect superior institutional resources, which should systematically translate into enhanced human capital accumulation and higher post-graduation compensation. Nevertheless, the postsecondary landscape in the United States is structurally heterogeneous, operating across distinct ownership classifications: public, private nonprofit, and for-profit institutions. These sectors differ fundamentally in their operational mandates, revenue reliance, pricing strategies, and admissions selectivity. Consequently, whether higher published tuition consistently delivers superior labor-market returns across all institutional sectors, or whether this cost-outcome relationship is moderated by institutional control, remains an unresolved empirical question with critical implications for educational equity and public policy.

To address this knowledge gap, the primary objective of this study is:

To determine whether a statistically significant relationship exists between published in-state tuition fees and median post-graduation earnings ten years post-entry, and to establish whether this tuition-earnings relationship differs systematically across public, private nonprofit, and for-profit postsecondary institutions.

Utilizing institutional-level data from the United States Department of Education College Scorecard across a cleaned sample of 2,894 postsecondary institutions, this research evaluates whether institutional pricing serves as a uniform signal of economic value or whether the marginal return on tuition is stratified by institutional sector.

To achieve the primary objective the study pursues four specific secondary objectives

- *Quantify Baseline Sectoral Stratification:*

  To measure and compare the baseline statistical distributions of published in-state tuition, 10-year post-graduation median earnings, cumulative graduate debt, and undergraduate enrollment across public, private nonprofit, and for-profit sectors, establishing sector-specific central tendencies and dispersion characteristics.

- *Evaluate Monotonic Bivariate Relationships:*

  To assess the functional form and statistical significance of the global and within-sector associations between in-state tuition and median earnings using non-parametric rank-order correlation analysis (Spearman's ρ), following formal diagnostic testing for distributional normality via the D'Agostino K^2 test.

- *Model Moderating Interaction Effects:*

  To construct sequential Ordinary Least Squares (OLS) regression models progressing from a bivariate baseline specification to sector-stratified interaction models and multivariable formulations in order to quantify how institutional control moderates the slope and intercept of the tuition-earnings return curve.

- *Assess Financial Mediation and Equity Implications:*

  To contextualize the cost-outcome returns by examining the mediating role of median graduate debt burdens across sectors and analyzing outcome disparities within Minority-Serving Institutions (MSIs) relative to non-MSIs.

2 Data Set and Variables

The empirical analysis evaluates institutional-level data drawn from the United States Department of Education College Scorecard, a public administrative database tracking published costs, student debt burdens, and post-graduation labor-market outcomes across higher education institutions in the United States. Mandatory federal reporting under Title IV student aid participation requirements establishes standardized variable definitions across institutional sectors.

Long-term economic outcomes are captured by matching de-identified federal tax records from the Department of the Treasury to former students ten years after their initial enrollment. The raw dataset comprises 6,273 postsecondary institutions evaluated across 24 variables capturing institutional identity, admissions selectivity, demographic designations, tuition pricing, graduate debt, and labor-market earnings. Nevertheless, missing data, non-numeric privacy suppressions, and extreme distribution tails necessitate systematic filtering to maintain statistical integrity. The raw sample was systematically refined through a multi-stage cleaning pipeline, resulting in a final analytical dataset of 2,894 postsecondary institutions. This sample reduction leads directly to the functional categorization of the variables evaluated in the study.

Table 1 provides a comprehensive inventory of all 24 variables present in the College Scorecard dataset, organized by operational function and analytical status. The core analytical framework relies on 10-year median earnings (`MD_EARN_WNE_P10`) as the primary outcome variable, published in-state tuition (`TUITIONFEE_IN`) as the primary predictor, and institutional control (`CONTROL`) as the categorical moderator across public ($N = 1,430$), private nonprofit ($N = 1,104$), and for-profit ($N = 360$) institutions. Cumulative graduate debt (`GRAD_DEBT_MDN`) and undergraduate enrollment (`UGDS`) serve as control covariates, while out-of-state tuition (`TUITIONFEE_OUT`) and predominant degree awarded (`PREDDEG`) provide baseline institutional context. Institutional identifiers, geographic location metrics, and two high-missingness admissions variables, namely average SAT scores (`SAT_AVG`) with 83.5 percent missingness and admission rates (`ADM_RATE`) with 69.6 percent missingness, were excluded from parametric regression modeling. Grouping these 24 variables by structural domain highlights the specific filtering protocols required to handle data anomalies across the sample.

Establishing valid parametric models requires explicit documentation of sample filtering rules, missing data handling, and distributional transformations. The 10-year median earnings metric includes only former students who received Title IV federal financial aid and whose tax records were matched by the Treasury Department, which may understate outcomes for non-aided higher-income graduates. String-based privacy suppressions in graduate debt ($N = 1,228$) were coerced to missing values prior to row deletion, while institutions charging zero tuition ($N = 1$) were removed to maintain a positive cost baseline. Undergraduate enrollment nulls ($N = 257$) were imputed using sector-specific medians to preserve scale differences across control types. Distributional distortion was mitigated by executing symmetric Interquartile Range trimming on 10-year earnings outside 11,818 dollars to 84,278 dollars, excluding 105 extreme tail observations, while tuition outliers were trimmed on a per-sector basis ($N = 62$). These rigorous cleaning protocols eliminate parametric distortion, providing an uncompromised analytical baseline for subsequent econometric modeling.

Table 1: Complete Inventory of College Scorecard Dataset Variables, Specifications, and Methodological Disclosures ($N = 24$)

Variable Name	Operational Description	Analytical Status and Methodological Disclosures
Core Analytical Variables		
MD_EARN_WNE_P10	Median post-graduation earnings 10 years post-entry in USD.	Target variable. Treasury tax-match data for working, non-enrolled Title IV aid recipients. 1,146 missing in raw. Symmetrically IQR-trimmed between 11,818 USD and 84,278 USD, removing 105 extreme outliers.
TUITIONFEE_IN	Published in-state tuition and fees in USD.	Primary predictor. 2,661 missing in raw. Zero-tuition records dropped ($N = 1$). Sector-stratified IQR-trimmed ($N = 62$ removed).
CONTROL	Ownership classification (1 Public, 2 Private Nonprofit, 3 For-Profit).	Primary moderator. Complete in raw data ($N = 6,273$). Encoded as CONTROL_LABEL for sector interaction modeling.
GRAD_DEBT_MDN	Median graduate cumulative debt in USD.	Mediating covariate. 1,228 PrivacySuppressed values coerced to missing, with 296 additional null rows removed.
UGDS	Total undergraduate degree-seeking enrollment.	Size control covariate. 795 missing in raw. 257 missing values in filtered subset imputed via per-sector medians.
Academic and Admissions Variables		
TUITIONFEE_OUT	Published out-of-state tuition and fees in USD.	Secondary cost variable. Retained for exploratory comparison, showing high correlation with in-state tuition in private sectors.
PREDDEG	Predominant degree awarded (0 Non-degree, 4 Graduate-focused).	Structural classification variable. Used to verify institutional degree-granting orientation across sectors.
SAT_AVG	Average SAT score of admitted students.	Excluded covariate. Excluded from regression models due to an extreme missingness rate of 83.5 percent ($N = 5,236$).

ADM_RATE	Proportion of undergraduate applicants admitted.	Excluded covariate. Excluded from regression models due to an extreme missingness rate of 69.6 percent ($N = 4,364$).
Institutional Identifiers and Geographic Indicators		
UNITID	Unique institution identifier assigned by Department of Education.	Primary key. Used for record tracking and duplicate validation, with zero duplicates found.
INSTNM	Official institution name.	Text identifier. Preserved for institutional lookup and reporting.
CITY	City in which institution is located.	Geographic identifier. Preserved for regional classification.
STABBR	Two-letter state abbreviation.	Geographic identifier. Preserved for state-level spatial analysis.
LOCALE	Urban-centric locale code (11 City Large, 43 Rural Remote).	Geographic classification. Captures campus urbanization and surrounding labor-market density.
Special Demographic and Institutional Designation Flags		
HBCU	Historically Black College or University flag (1 Yes).	MSI Indicator. Aggregated into composite MSI variable for sectoral equity analysis.
PBI	Predominantly Black Institution flag (1 Yes).	MSI Indicator. Aggregated into composite MSI variable for sectoral equity analysis.
ANNHI	Alaska Native or Native Hawaiian Serving Institution flag (1 Yes).	MSI Indicator. Aggregated into composite MSI variable for sectoral equity analysis.
TRIBAL	Tribal College or University flag (1 Yes).	MSI Indicator. Aggregated into composite MSI variable for sectoral equity analysis.
AANAPII	Asian American Native American Pacific Islander Serving flag (1 Yes).	MSI Indicator. Aggregated into composite MSI variable for sectoral equity analysis.
HSI	Hispanic-Serving Institution flag (1 Yes).	MSI Indicator. Aggregated into composite MSI variable for sectoral equity analysis.
NANTI	Native American Non-Tribal Institution flag (1 Yes).	MSI Indicator. Aggregated into composite MSI variable for sectoral equity analysis.
MENONLY	Men-only institution flag (1 Yes).	Single-sex designation. Evaluated for institutional baseline characteristics.
WOMENONLY	Women-only institution flag (1 Yes).	Single-sex designation. Evaluated for institutional baseline characteristics.
RELAFFIL	Religious affiliation code.	Cultural or institutional flag. Captures private institutional religious governance.

3 Data Cleaning & Feature Engineering

The raw College Scorecard dataset contains 6,273 postsecondary institutions evaluated across 24 institutional attributes. Establishing a valid analytical sample requires systematic filtering of incomplete variables to prevent parametric estimation bias (Pepinsky, 2018). Two admissions variables exhibited extreme missingness, with average SAT scores missing in 83.5 percent of institutions ($N = 5,236$) and admission rates missing in 69.6 percent of cases ($N = 4,364$). Imputing values across a majority of unobserved records would introduce unacceptable selection bias. Furthermore, key outcome and predictor variables contained substantial missing data, including 2,661 missing entries in published in-state tuition and 1,146 missing entries in 10-year median earnings. Highly incomplete admissions variables were excluded from parametric regression modeling, while rows lacking primary target or predictor values were systematically removed from the analytical sample (Pepinsky, 2018).

Standardizing numerical data types is essential for maintaining computational integrity across financial metrics. The median graduate debt column contained 1,228 non-numeric PrivacySuppressed string entries, which the Department of Education inserts when cell counts fall below reporting thresholds. These string entries were systematically coerced to missing numeric values (Pepinsky, 2018), increasing the total null count for graduate debt to 1,496 rows. Subsequently, sequential row-wise exclusions were executed by dropping 2,915 institutions missing either tuition or 10-year earnings, 1 institution reporting zero in-state tuition, and 296 remaining institutions missing graduate debt data. Eliminating string artifacts and zero-cost entries reduced the dataset to 3,061 observations, establishing a positive baseline cost structure required for investment-return modeling (Muse & Muse, 2024).

Addressing missing values in control covariates requires imputation strategies that preserve underlying sectoral differences. Undergraduate enrollment contained 257 missing values within the filtered sample of 3,061 institutions. Applying a global median imputation would obscure the vast scale disparities between large public universities and smaller private entities.

Missing enrollment values were therefore imputed using sector-specific medians derived within each institutional control classification (Faber, 2017). Following imputation, primary key validation was conducted on unique unit identifiers, confirming zero duplicate institutional records across the dataset. Sector-stratified imputation preserved structural scale variance, leading directly to the outlier detection phase.

Trimming distribution tails prevents extreme values from distorting regression slope estimates. The 10-year median earnings distribution exhibited right-tail skewness and a floor effect among low-earning institutions. Applying symmetric Interquartile Range trimming between 11,818 dollars and 84,278 dollars identified and removed 105 extreme earnings outliers (Muse & Muse, 2024). In-state tuition distributions differed dramatically by sector, rendering global Interquartile Range bounds inappropriate for cost filtering. Consequently, tuition outliers were trimmed independently within each institutional control sector (Faber & Slantcheva-Durst, 2021), removing 62 additional institutions. Dual-stage outlier trimming eliminated extreme leverage points, producing a final analytical dataset of 2,894 postsecondary institutions.

Feature engineering transforms raw administrative codes into analytically meaningful categorical predictors and composite indicators. The numeric ownership classification was converted into a human-readable categorical sector variable, partitioning the final sample into 1,430 public institutions, 1,104 private nonprofit institutions, and 360 for-profit institutions (Faber & Slantcheva-Durst, 2021). Seven individual minority-serving institution flags, encompassing Historically Black Colleges, Predominantly Black Institutions, Alaska Native or Native Hawaiian Serving Institutions, Tribal Colleges, Asian American Native American Pacific Islander Serving Institutions, Hispanic-Serving Institutions, and Native American Non-Tribal Institutions, were aggregated into a unified binary indicator distinguishing minority-serving institutions from non-minority-serving institutions (Faber & Slantcheva-Durst, 2021). Constructing sector labels and composite equity indicators enables the subsequent econometric modeling of interaction effects and baseline outcome disparities.

Table 2 summarizes the step-by-step reduction of the sample across each stage of the data cleaning pipeline.

Table 2: Sequential Data Cleaning and Sample Selection Pipeline

Pipeline Stage	**Observations Removed**	**Remaining Sample Size**
Raw College Scorecard Dataset	0	6,273
Missing Target or Predictor (`MD_EARN_WNE_P10` or `TUITIONFEE_IN`)	2,915	3,358
Zero In-State Tuition (`TUITIONFEE_IN` = 0)	1	3,357
Missing Graduate Debt (`GRAD_DEBT_MDN` after coercion)	296	3,061
Duplicate Institution Check (`UNITID`)	0	3,061
Target Earnings Outliers (Global IQR Bounds)	105	2,956
Predictor Tuition Outliers (Sector-Stratified IQR Bounds)	62	2,894
Final Clean Analytical Sample	**3,379**	**2,894**

4 Descriptive Statistics

Evaluating institutional control distributions establishes the baseline structural composition of the final analytical sample. Public postsecondary institutions constitute the largest single sector within the cleaned dataset ($N = 1,430$, representing 49.4 percent of the sample), followed by private nonprofit institutions ($N = 1,104$, or 38.1 percent) and for-profit institutions ($N = 360$, or 12.4 percent). Public institutions dominate total undergraduate enrollment with a mean of 6,758 students and a median of 3,953 students, whereas private nonprofit institutions maintain a median enrollment of 1,223 students and for-profit institutions operate at a median scale of 396 students. Table 3 details the institutional distribution across ownership classifications. Sectoral distribution highlights a market dominated by public and private nonprofit institutions, leading directly to an analysis of baseline pricing and earnings disparities across ownership types.

Table 3: Institutional Sample Distribution by Ownership Classification

Institutional Sector	**Institution Count (N)**	**Sample Percentage (%)**
Public	1,430	49.4%
Private Nonprofit	1,104	38.1%
For-Profit	360	12.4%
Total Clean Sample	**2,894**	**100.0%**

Published in-state tuition fees reflect fundamental differences in funding mechanisms and administrative mandates across institutional sectors. Private nonprofit institutions report the highest published tuition fees, exhibiting a mean of 33,370 dollars and a median of 33,928 dollars, alongside a broad interquartile range of 23,741 dollars. Conversely, public universities charge a mean tuition of 6,786 dollars and a median of 5,981 dollars, operating at roughly one-fifth the published price of private nonprofits. For-profit institutions occupy an intermediate pricing tier, with a mean tuition of 15,934 dollars, a median of 16,326 dollars, and a compact interquartile range of 4,225 dollars. Overall published tuition averages 18,065 dollars across the entire sample with a median of 12,221 dollars. Disparities in tuition pricing demonstrate substantial cost asymmetry, setting the stage for evaluating post-graduation earnings returns.

Labor-market returns ten years post-entry exhibit sector-specific central tendencies and outcome dispersion. Private nonprofit graduates achieve the highest labor-market compensation, recording a mean median earnings of 52,375 dollars and a median of 52,198 dollars, accompanied by a standard deviation of 13,368 dollars. Public university graduates achieve highly competitive economic outcomes despite lower initial tuition, generating a mean median earnings of 45,993 dollars and a median of 43,466 dollars. For-profit institution graduates trail both sectors significantly, reporting a mean median earnings of 38,274 dollars and a median of 37,480 dollars. Overall median earnings across all 2,894 institutions average 47,468 dollars with a median of 45,378 dollars. Earnings stratification confirms that higher tuition does not guarantee superior earnings across all sectors, prompting a cross-metric evaluation of graduate debt burdens.

Cumulative graduate debt serves as a mediating financial variable that qualifies the net economic return of higher education. Private nonprofit graduates accumulate the highest median debt burdens, averaging 22,974 dollars with a median of 24,517 dollars. For-profit graduates carry a mean debt of 18,876 dollars and a median debt of 18,668 dollars, representing a severe debt burden relative to their lower 10-year median earnings of 37,480 dollars. Public university graduates experience the lowest debt accumulation, with a mean of 14,594 dollars and a median of 12,250 dollars. Overall graduate debt averages 18,324 dollars across all institutions with a median of 19,500 dollars. Table 4 presents the complete multi-variable descriptive statistics split by institutional control and overall sample totals. Triangulating tuition pricing, graduate earnings, and debt accumulation reveals that public institutions deliver superior financial efficiency, whereas for-profit institutions generate the highest structural debt-to-earnings risk.

Table 4: Descriptive Statistics for Postsecondary Cost, Earnings, Debt, and Enrollment by Sector

Variable and Metric	Sectoral Breakdown (Public / Private Nonprofit / For-Profit)	Overall Sample Total
In-State Tuition		
Mean	\$6,786 / \$33,370 / \$15,934	\$18,065
Median	\$5,981 / \$33,928 / \$16,326	\$12,221
Standard Deviation	\$3,572 / \$16,069 / \$3,626	\$16,101
Interquartile Range	\$4,664 / \$23,741 / \$4,225	\$19,101
10-Year Median Earnings		
Mean	\$45,993 / \$52,375 / \$38,274	\$47,468
Median	\$43,466 / \$52,198 / \$37,480	\$45,378
Standard Deviation	\$10,269 / \$13,368 / \$10,132	\$12,400
Interquartile Range	\$12,918 / \$17,056 / \$11,419	\$16,696
Median Graduate Debt		
Mean	\$14,594 / \$22,974 / \$18,876	\$18,324
Median	\$12,250 / \$24,517 / \$18,668	\$19,500
Standard Deviation	\$6,148 / \$5,087 / \$9,792	\$7,457
Interquartile Range	\$10,387 / \$4,948 / \$13,743	\$13,684
Undergraduate Enrollment		
Mean	6,758 / 2,196 / 1,355	4,346
Median	3,953 / 1,223 / 396	1,792
Standard Deviation	8,294 / 7,657 / 5,795	8,140
Interquartile Range	6,056 / 1,356 / 310	3,454

5 Methodology

5.1 Data Source and Sample

The analysis uses a subset of the United States Department of Education College Scorecard, a public database that reports institutional characteristics, costs, student outcomes, and debt for postsecondary institutions across the country. The Scorecard was developed to help students and families make informed decisions about higher education by providing transparent and comparable data on costs and expected earnings.

The data are collected through federal reporting requirements. Institutions that participate in Title IV federal student aid programs must report enrollment, completion, tuition, and financial aid data to the Department of Education. Earnings outcomes are drawn from de-identified tax records matched to former students through the Treasury Department, which provides a reliable measure of labor-market returns. The collection process imposes standardized definitions across institutions, which strengthens internal comparability. However, the earnings figures are limited to students who received federal financial aid and whose tax records could be matched. This means the earnings data may not represent all graduates and may understate outcomes for students from higher-income families who did not receive aid.

The dataset contains 6,273 rows and 24 columns. Each row represents one postsecondary institution identified by a unique unit identifier. The columns capture institutional identity, admissions selectivity, demographic designations, tuition, student debt, and post-graduation earnings. The key variables for this study are in-state tuition (TUITIONFEE_IN), median earnings ten years after entry (MD_EARN_WNE_P10), institutional control type (CONTROL), median graduate debt (GRAD_DEBT_MDN), and undergraduate enrollment (UGDS). The full list of variables is available in the College Scorecard data dictionary.

5.2 Data Cleaning

The raw dataset required substantial cleaning before analysis. Five variables were selected as central to the research question: in-state tuition, median post-graduation earnings, median graduate debt, institutional control type, and undergraduate enrollment. Two variables were excluded from the primary analysis. SAT average scores were missing for 83.5% of institutions and admission rates for 69.6%. Imputing values for the majority of observations would introduce bias and undermine the validity of downstream inferences. Institutions reporting zero in-state tuition were also removed, because the investment-return framing of this study requires a positive cost baseline against which to measure earnings.

The cleaning pipeline addressed six issues sequentially. First, the graduate debt column contained "PrivacySuppressed" string values for 1,228 institutions. These were coerced to numeric NaN using `pd.to_numeric` with `errors='coerce'`. Second, rows missing the target variable (MD_EARN_WNE_P10) or the primary predictor (TUITIONFEE_IN) were dropped. Third, rows missing graduate debt after the privacy suppression conversion were dropped. Fourth, missing undergraduate enrollment values were imputed with the median enrollment within each sector rather than a global median, because enrollment distributions differ substantially across public, private nonprofit, and for-profit institutions. Fifth, duplicate UNITID values were detected and removed. Sixth, outliers were trimmed using the interquartile range method. Earnings outliers were removed symmetrically across the full sample. Tuition outliers were removed within each sector separately, because tuition distributions differ dramatically by control type and a global IQR filter would disproportionately remove legitimate observations from the private nonprofit tier.

The final cleaning step created a human-readable sector label from the numeric CONTROL code, mapping 1 to Public, 2 to Private Nonprofit, and 3 to For-Profit. The pipeline reduced the dataset from 6,273 to 2,894 observations. The largest reduction came from the 2,915 institutions that lacked valid tuition or earnings data. The median imputation for undergraduate enrollment retained 257 institutions that would otherwise have been dropped, and the sector-specific IQR outlier removal stripped 167 extreme values. The final dataset contains no missing values in any of the five target variables.

5.3 Descriptive Framework

Summary statistics were computed stratified by institutional control type. Panel A reports the sector composition of the cleaned dataset. The final sample of 2,894 institutions comprises 1,430 public institutions (49.4

The descriptive statistics serve two purposes. They establish the central tendencies and dispersions that any model of the tuition-earnings relationship must account for. They also reveal the sector-level disparities that motivate the stratified modeling approach. If earnings and tuition differ systematically by sector, a model that pools all institutions into a single slope will misrepresent the relationship for every sector.

5.4 Exploratory Data Analysis

Six exploratory questions were posed, moving from univariate description to joint and contextual relationships. The first question examined the distribution of post-graduation earnings across sectors using grouped summary statistics, kernel density estimation, and boxplots. The second examined in-state tuition distributions using grouped statistics and violin plots, with the national median as a benchmark. The third assessed the joint tuition-earnings relationship through Spearman rank-order correlations (Hotelling & Pabst, 1936) and scatter plots with sector-stratified regression lines. The fourth examined graduate debt as a mediating variable, computing Spearman correlations between tuition and debt alongside median debt-to-earnings ratios for each sector (Faber & Slantcheva-Durst, 2021). The fifth tested whether the tuition-earnings link persisted within admission selectivity tiers, stratifying institutions into highly selective (admission rate below 40%), moderately selective (40% to 70%), and less selective (above 70%) categories (Dale & Krueger, 2011; Muse & Muse, 2024). The sixth compared minority-serving institutions against non-minority-serving institutions within each sector using grouped means and medians for tuition, earnings, and debt (Faber & Slantcheva-Durst, 2021).

These exploratory analyses established three patterns that directly informed the modeling strategy. Earnings and tuition differ systematically by sector. The within-sector tuition-earnings correlation is strongest among public institutions and weakest among for-profits. And the debt-to-earnings ratio is substantially lower for public graduates than for graduates of private nonprofit or for-profit institutions.

5.5 Statistical Inference Framework

Before estimating the regression models, the distribution of median earnings was tested for normality using the D'Agostino K-squared test. The null hypothesis was that the earnings data follow a normal distribution. The alternative was that they deviate from normality. Because the earnings distribution was expected to show positive skew and multimodality given the sector-level differences observed in the exploratory analysis, a significant test result would justify the use of Spearman's rank correlation for the preliminary association test (Hotelling & Pabst, 1936). The significance level was set at $\alpha = 0.05$.

The D'Agostino K-squared test produces a test statistic and p-value, and if the data deviate significantly from normality, the analysis uses Spearman's rank correlation rather than Pearson's r (Hotelling & Pabst, 1936). Visual diagnostics including a histogram and Q-Q plot supplement the formal test to assess whether deviations from normality are practically severe or merely a consequence of the large sample size. The Spearman correlation coefficient and its associated p-value are reported in the Results section.

5.6 Regression Modeling Strategy

The research question asks whether a significant relationship exists between in-state tuition and median post-graduation earnings and whether this relationship differs across public, private nonprofit, and for-profit institutions. A nested sequence of three ordinary least squares models was specified to answer both parts of the question. This design follows the institution-level higher-education research literature. Faber (2017, pp. 41–45) and Faber and Slantcheva-Durst (2021, pp. 687–700) model earnings using College Scorecard institutional attributes including state, enrollment, and tuition. Muse and Muse (2024, pp. 36–40) likewise model College Scorecard earnings with tuition, institutional type, size, degree offerings, geography, and selectivity controls.

All three models use the natural logarithm of in-state tuition and the natural logarithm of median post-graduation earnings. Both variables are positive and right-skewed, and the log transformation reduces skewness while producing coefficients that are interpretable as elasticities. A coefficient of 0.10 means that a 1% increase in tuition is associated with approximately 0.10% higher earnings. Muse and Muse (2024, pp. 37, 40) apply the same logarithmic transformation to College Scorecard median earnings. Undergraduate enrollment enters as $\log(1 + \text{UGDS})$ because one institution reports zero enrollment, and the log of zero is undefined.

All models use HC3 heteroskedasticity-consistent standard errors. MacKinnon and White (1985, pp. 307–309) derive the jackknife-based HC3 covariance estimator for regression inference under heteroskedasticity, and the Breusch-Pagan test for constant residual variance is reported for each model. The estimates describe conditional associations and must not be interpreted as causal effects of tuition on earnings.

5.6.1 Model 1: Unconditional Association

Model 1 estimates the overall association between in-state tuition and median earnings without adjusting for institutional characteristics. The specification is

$$\log(\text{Earnings})_i = \beta_0 + \beta_1 \log(\text{Tuition})_i + \varepsilon_i$$

where i indexes institutions. The coefficient β_1 represents the pooled tuition elasticity across all sectors. This model addresses the first part of the research question but pools public, private nonprofit, and for-profit institutions into a single slope, which the exploratory analysis has shown to be misleading.

5.6.2 Model 2: Sector Interaction Model

Model 2 adds institutional sector indicators and interacts each sector with log tuition. The specification is

$$\begin{aligned}\log(\text{Earnings})_i = \beta_0 &+ \beta_1 \log(\text{Tuition})_i \\ &+ \beta_2 \text{Nonprofit}_i + \beta_3 \text{ForProfit}_i \\ &+ \beta_4 \log(\text{Tuition})_i \times \text{Nonprofit}_i \\ &+ \beta_5 \log(\text{Tuition})_i \times \text{ForProfit}_i + \varepsilon_i\end{aligned}$$

Public institutions are the reference category. The coefficients β_4 and β_5 test whether the tuition elasticity for private nonprofit and for-profit institutions differs from the public institution elasticity. A joint HC3 Wald test evaluates whether the two interaction coefficients are jointly zero, which directly answers the second part of the research question. Muse and Muse (2024, pp. 36, 39) distinguish the same three control types when modeling College Scorecard earnings, and Faber and Slantcheva-Durst (2021, pp. 687–700) show that institutional attributes change the interpretation of tuition-earnings relationships.

5.6.3 Model 3: Institutional-Control Robustness Model

Model 3 retains the Model 2 interaction terms and adds three prespecified blocks of institutional controls. The specification is

$$
\begin{aligned}
\log(\text{Earnings})_i = \beta_0 &+ \beta_1 \log(\text{Tuition})_i \\
&+ \beta_2 \text{Nonprofit}_i + \beta_3 \text{ForProfit}_i \\
&+ \beta_4 \log(\text{Tuition})_i \times \text{Nonprofit}_i \\
&+ \beta_5 \log(\text{Tuition})_i \times \text{ForProfit}_i \\
&+ \beta_6 \log(1 + \text{Enrollment})_i \\
&+ \gamma \cdot \text{Degree}_i + \delta \cdot \text{State}_i + \varepsilon_i
\end{aligned}
$$

Undergraduate enrollment enters as $\log(1 + \text{UGDS})$. Predominant degree enters categorically with bachelor's degree as the reference category, using the PREDDEG variable where 0 indicates non-degree-granting, 1 indicates certificate or associate, 2 indicates associate, 3 indicates bachelor's, and 4 indicates graduate-focused. Institution state enters as fixed effects using the STABBR variable, with Guam, Puerto Rico, and the U.S. Virgin Islands grouped as "US Territory" to avoid separate fixed effects supported by only one or two observations.

The control set follows the closest published precedents that can be mapped to this dataset. Faber (2017, pp. 42–45) includes state, enrollment, and tuition among blocked College Scorecard institutional attributes before estimating earnings with OLS. Muse and Muse (2024, pp. 36–40) include selectivity, region, institutional type, size, degree offerings, tuition, and log earnings. Graduate debt is excluded from the primary model because it may lie on the causal pathway from tuition to later earnings outcomes. Admission rate is handled separately in a disclosed sensitivity analysis because it is observed for only 50.7% of the cleaned sample.

The robustness question is whether the sector-specific tuition elasticities remain positive and whether the interaction terms remain jointly significant after these controls. A joint HC3 Wald test evaluates whether the additional institutional controls contribute explanatory information beyond Model 2.

5.6.4 Admission-Rate Sensitivity Analysis

Admission rate is added only as a disclosed sensitivity analysis and does not replace the full-sample Model 3. Muse and Muse (2024, pp. 35–37) treat admission rate as a selectivity indicator but document that selectivity measures are missing for many institutions, and the same problem is material here. Only 1,466 of 2,894 institutions report ADM_RATE, and the reporting subset contains 908 private nonprofits, 503 public institutions, and only 55 for-profits.

Two models are estimated on exactly that restricted sample. The first repeats the Model 3 specification without admission rate. The second adds admission rate as a linear predictor. Comparing the two separates the effect of adding selectivity from the effect of changing the estimation sample. Missing admission rates are unlikely to be random, because selective institutions are disproportionately likely to report their rates. The restricted results must not be generalized to all institutions.

5.7 Hypothesis Testing and Inference

All hypothesis tests use a significance level of $\alpha = 0.05$. The null hypothesis for the overall tuition-earnings association is $\mathrm{H}_0 : \beta_1 = 0$ against $\mathrm{H}_1 : \beta_1 \neq 0$. The null hypothesis for the sector differences is that the tuition-by-sector interaction coefficients are jointly zero, tested using an HC3 Wald statistic.

Model fit is assessed using adjusted R-squared, log RMSE, AIC, and BIC. The Breusch-Pagan test for heteroskedasticity is reported for each model to validate the use of HC3 standard errors. The sector-specific tuition elasticities and their 95% confidence intervals are computed using the delta method via linear hypothesis tests on the combined coefficients. For each sector, the elasticity is the sum of β_1 and the relevant interaction term, and the standard error is computed from the HC3 covariance matrix.

5.8 Software

All analyses were conducted in Python 3.12. Data manipulation used pandas (version 3.0) and numpy (version 2.5). Normality testing and correlation analysis used SciPy (version 1.18). Regression models were estimated using statsmodels (version 0.14) with the `smf.ols` interface and HC3 covariance specification. Visualizations were generated using matplotlib (version 3.11) and seaborn (version 0.13), styled following the Economist-aligned Quantitative Visualization and Reporting Standard. The complete analysis code is available in the companion Jupyter notebook.

6 Results

6.1 Sample Composition and Descriptive Patterns

The cleaning pipeline reduced the College Scorecard dataset from 6,273 to 2,894 institutions. The final sample contains 1,430 public institutions (49.4%), 1,104 private nonprofit institutions (38.1%), and 360 for-profit institutions (12.4%). Table 5 reports the sector composition and summary statistics for the four continuous variables.

Private nonprofit institutions charge the highest tuition and produce the highest median earnings. Public institutions charge roughly one-fifth of the nonprofit tuition level yet produce competitive earnings. For-profit institutions occupy a middle tuition tier but deliver the lowest earnings. The asymmetry is the key descriptive fact: the market does not price education uniformly, and it does not reward tuition spending proportionally.

6.2 Exploratory Patterns

Six exploratory questions guided the analysis before model estimation. The answers to each question established the context for the regression models and revealed the sector-level patterns that the research question must address.

Table 5: Descriptive Statistics by Institutional Control Type

Panel A: Sector Composition

Sector	Count	Percent
Public	1,430	49.4
Private Nonprofit	1,104	38.1
For-Profit	360	12.4
Total	2,894	100.0

Panel B: Summary Statistics by Sector

Variable	Statistic	Public	Nonprofit	For-Profit	Overall
In-State Tuition	Mean	$6,786	$33,370	$15,934	$18,065
	Median	$5,981	$33,928	$16,326	$12,221
	Std Dev	$3,572	$16,069	$3,626	$16,101
	IQR	$4,664	$23,741	$4,225	$19,101
Median Earnings	Mean	$45,993	$52,375	$38,274	$47,468
	Median	$43,466	$52,198	$37,480	$45,378
	Std Dev	$10,269	$13,368	$10,132	$12,400
	IQR	$12,918	$17,056	$11,419	$16,696
Median Graduate Debt	Mean	$14,594	$22,974	$18,876	$18,324
	Median	$12,250	$24,517	$18,668	$19,500
	Std Dev	$6,148	$5,087	$9,792	$7,457
	IQR	$10,387	$4,948	$13,743	$13,684
Undergraduate Enrollment	Mean	6,758	2,196	1,355	4,346
	Median	3,953	1,223	396	1,792
	Std Dev	8,294	7,657	5,795	8,140
	IQR	6,056	1,356	310	3,454

Note. N = 2,894 institutions. Earnings are median values ten years after entry. Debt is the median cumulative debt of graduates. IQR is the interquartile range.

6.2.1 Question 1: How do earnings vary by sector?

Figure 1 shows the distribution of post-graduation earnings across the three sectors. Private nonprofit earnings are the highest on average but also the most dispersed, with an interquartile range of $17,056. Public institutions show the tightest clustering around the central tendency: the kernel density peak is narrow and tall, reflecting a consistent earnings band of roughly $40,000 to $50,000 for most public graduates. For-profit earnings are concentrated at the low end of the distribution, with a median of $37,480 that falls below the first quartile of the public sector.

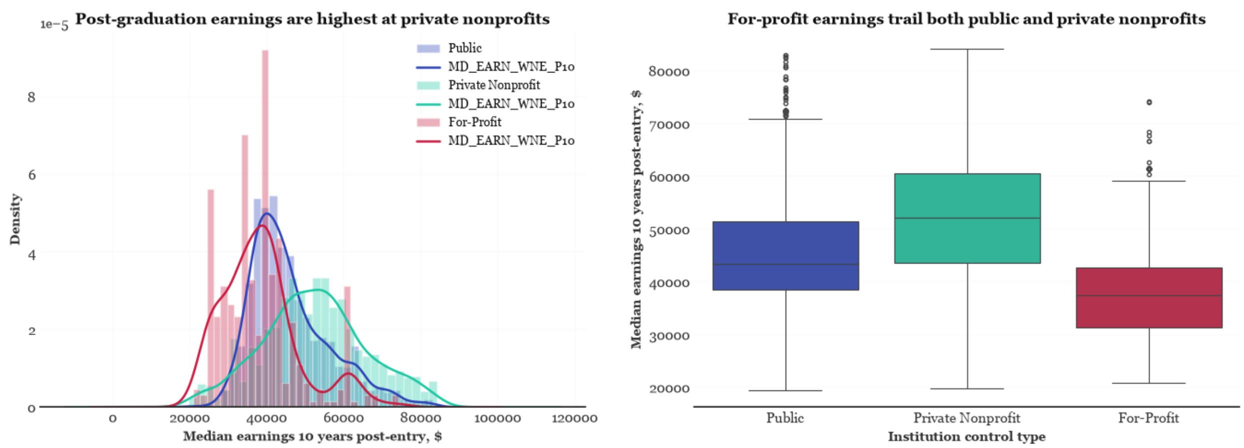

Figure 1: Distribution of median post-graduation earnings by institutional sector. Left panel shows kernel density estimates. Right panel shows boxplots. Private nonprofit graduates show the highest but most variable earnings. For-profit graduates are concentrated at the lowest levels.

6.2.2 Question 2: How does tuition vary by sector?

Tuition distributions are more sharply separated than earnings distributions. Figure 2 reveals three distinct pricing regimes. Public institutions charge a median of $5,981, and fewer than 10% exceed the national median of $12,221. Private nonprofits charge a median of $33,928, nearly three times the national median, with 91.2% of institutions pricing above it. For-profit institutions charge a median of $16,326, which falls halfway between the public and nonprofit levels, yet 84.4% still charge above the national median. The contrast with the earnings distribution is stark: public institutions produce competitive earnings from roughly one-fifth the tuition base of private nonprofits.

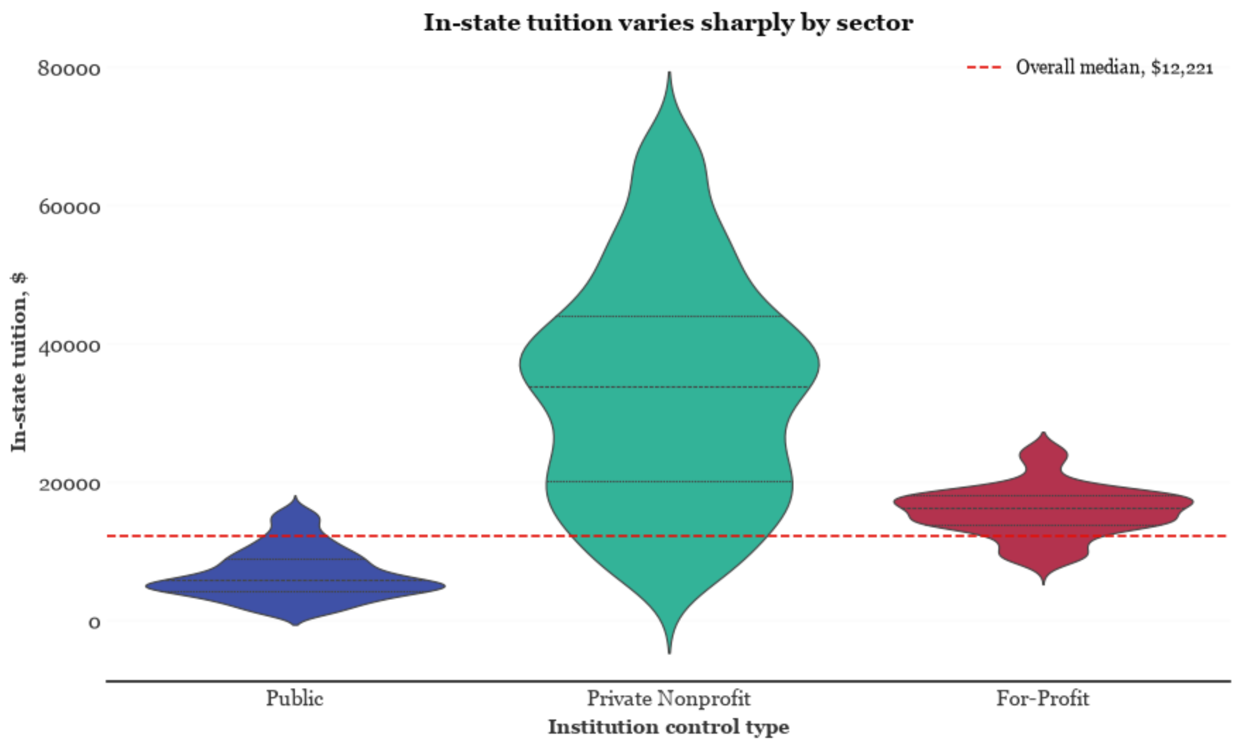

Figure 2: In-state tuition distribution by sector with the national median ($12,221) marked as a dashed red line. Public tuition is compressed in a low band. Private nonprofit tuition spans an extremely wide range. For-profit tuition is tightly clustered above the national median.

6.2.3 Question 3: Do tuition and earnings form distinct clusters by sector?

Figure 3 presents the joint relationship. The three sectors form distinct clusters rather than a single continuum. Public institutions occupy a compact region of low tuition and moderate earnings. Private nonprofits stretch across a wide tuition range with highly variable earnings. For-profit institutions cluster in a narrow tuition band but show the lowest earnings conditional on tuition.

The overall Spearman rank correlation between tuition and earnings is 0.464 ($p < .001$), confirming a statistically significant positive monotonic relationship. Stratified by sector, the correlations diverge considerably. Public institutions show the strongest association at 0.629. Private nonprofits follow at 0.587. For-profit institutions show a substantially weaker correlation of 0.335, indicating that tuition is a far less reliable signal of future earnings in that sector.

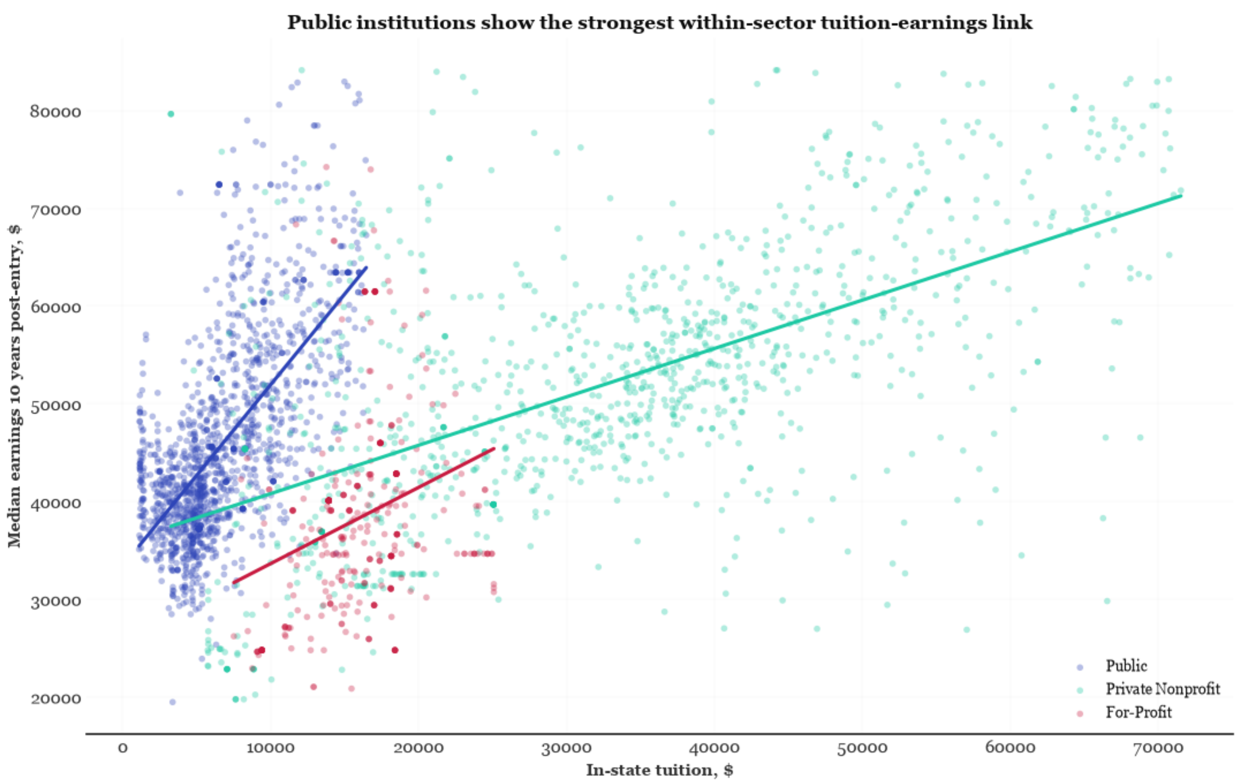

Figure 3: Tuition and earnings by institutional sector with linear regression lines. The three sectors form distinct clusters with different slopes and intercepts. The for-profit cloud sits below the other two at every tuition level.

6.2.4 Question 4: How does graduate debt relate to tuition and earnings?

Graduate debt complicates the simple tuition-earnings story. Figure 4 shows two patterns. First, tuition and debt correlate positively in all three sectors. The correlation is strongest in the public sector (Spearman r = 0.729), reflecting a direct pass-through of costs to students. Private nonprofits show a ceiling effect where debt plateaus around $25,000 despite tuition reaching $80,000, consistent with federal borrowing limits constraining debt even at high tuition levels.

Second, the debt-to-earnings ratio reveals the real divide. Public graduates carry a median ratio of 0.29, roughly twenty-nine cents of debt for every dollar of annual earnings. Private nonprofit and for-profit graduates carry ratios of 0.45 and 0.43, respectively. The cost of financing education through debt erodes the net return, and it does so most severely outside the public sector. This finding adds weight to the research question: if students in different sectors face systematically different debt burdens, the net benefit of a tuition investment cannot be evaluated on earnings alone.

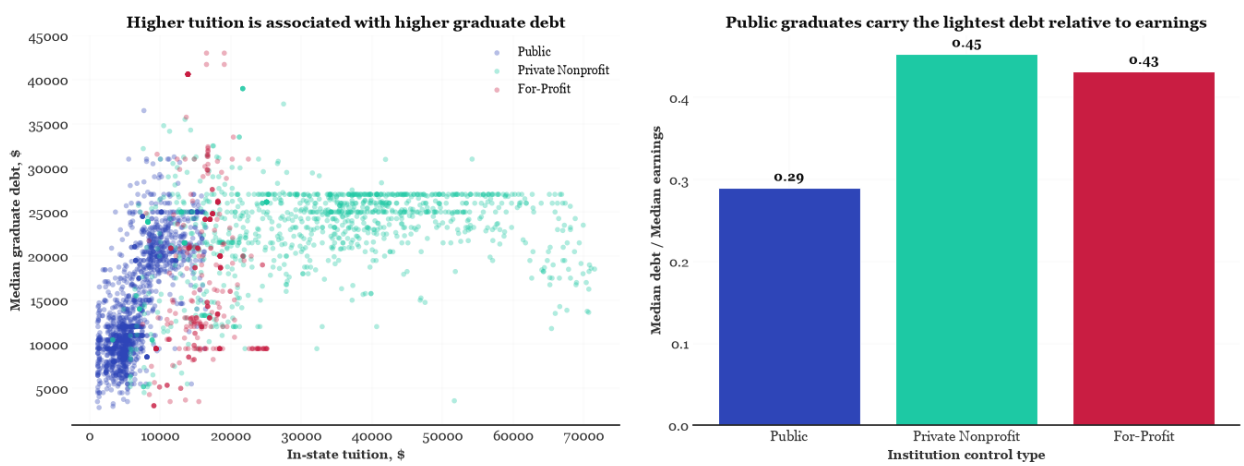

Figure 4: Left panel: tuition versus graduate debt by sector. Right panel: median debt-to-earnings ratio by sector. Public graduates carry the lightest debt burden relative to their earnings.

6.2.5 Question 5: Does selectivity explain the tuition-earnings link?

Admission rate data are available for roughly half the sample, so this question is examined as a stratified analysis rather than a control variable. Figure 5 shows a clear selectivity gradient:

more selective institutions produce higher earnings in every sector. Among highly selective institutions, private nonprofit graduates earn a median of $63,952 compared to $54,064 in the less selective tier.

The tuition-earnings correlation persists within selectivity tiers for public and private nonprofit institutions. Public institutions maintain a significant Spearman correlation of 0.404 within the reporting subset, and private nonprofits show a correlation of 0.551. The for-profit sector produces a negative but unreliable estimate in this restricted sample, driven by the small number of for-profit institutions that report admission rates. Selectivity therefore explains part of the earnings variation but does not eliminate the sector-level pattern.

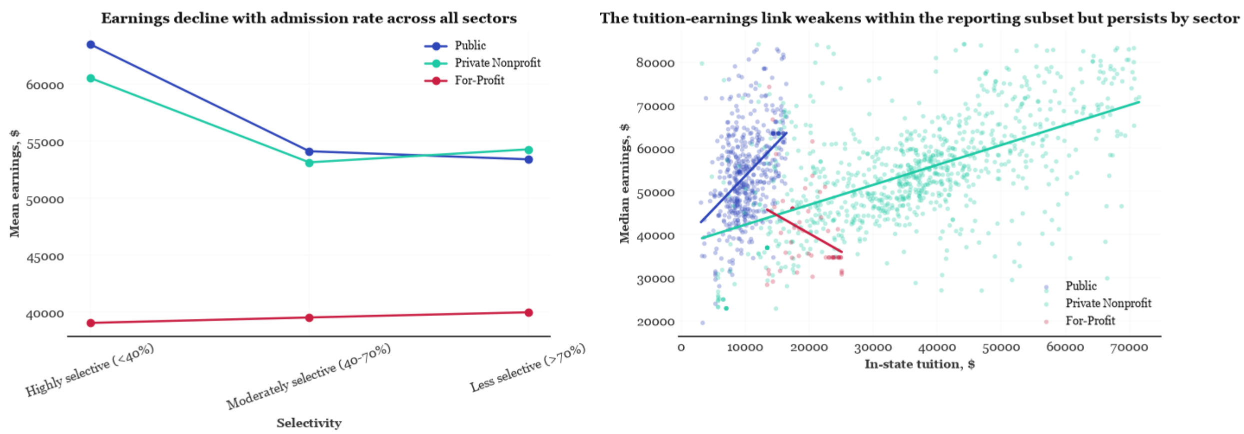

Figure 5: Left panel: mean earnings by selectivity tier and sector. Right panel: tuition-earnings scatter within the admission-reporting subset with regression lines. The sector gradient persists after accounting for selectivity.

6.2.6 Question 6: Do minority-serving institutions differ from non-MSIs?

Minority-serving institutions operate on a structurally different cost-outcome curve within every sector. Figure 6 shows that MSIs charge lower median tuition and report lower median earnings than their non-MSI counterparts. The gap is most visible in the public sector, where MSIs charge a median of $5,193 against $6,361 for non-MSIs, and their graduates earn a median of $43,172 against $44,298.

Public MSIs also generate lower median graduate debt ($11,631 versus $13,532). These institutions succeed in minimizing front-end costs and debt burdens. The persistently lower earnings, however, point to an equity dimension: these institutions serve populations that face broader socioeconomic and labor market barriers that the tuition-earnings models cannot capture. This finding does not directly test the research question, but it adds context: if the tuition-earnings relationship differs by sector, and MSIs within each sector sit at the lower end of both distributions, the net return equation is not uniform even within sectors.

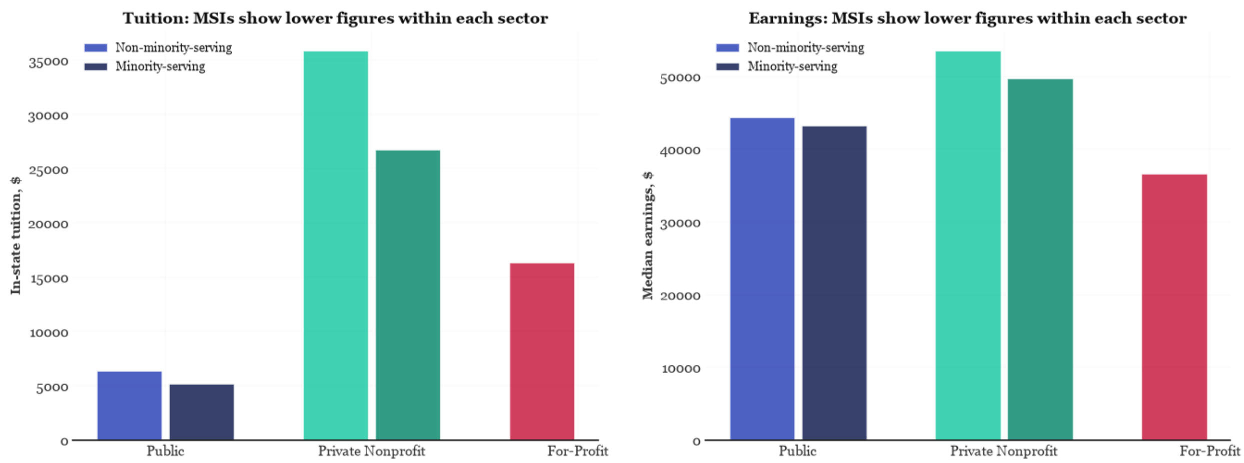

Figure 6: Median tuition and median earnings by sector, comparing minority-serving institutions against non-minority-serving institutions. MSIs charge less and their graduates earn less within every sector.

6.3 Preliminary Inference

Before estimating the parametric regression models, the 10-year median earnings distribution was formally tested for normality. Because the earnings data significantly deviated from normality ($p < 0.001$), a non-parametric Spearman rank-order correlation was performed. The D'Agostino K-squared test confirmed that the distribution deviates significantly from a Gaussian curve ($p < .001$). The visual diagnostics in Figure 7 supplement this formal test: the histogram reveals a mixture distribution with a primary mode near $42,000 driven by the public sector, a right shoulder near $60,000 reflecting private nonprofits, and a left tail near $20,000 capturing for-profit institutions, while the Q-Q plot shows lighter tails at both extremes.

The non-parametric correlation test revealed a moderate, positive correlation between published in-state tuition and 10-year median post-graduation earnings ($r_s = 0.4641$, $p < 0.001$).

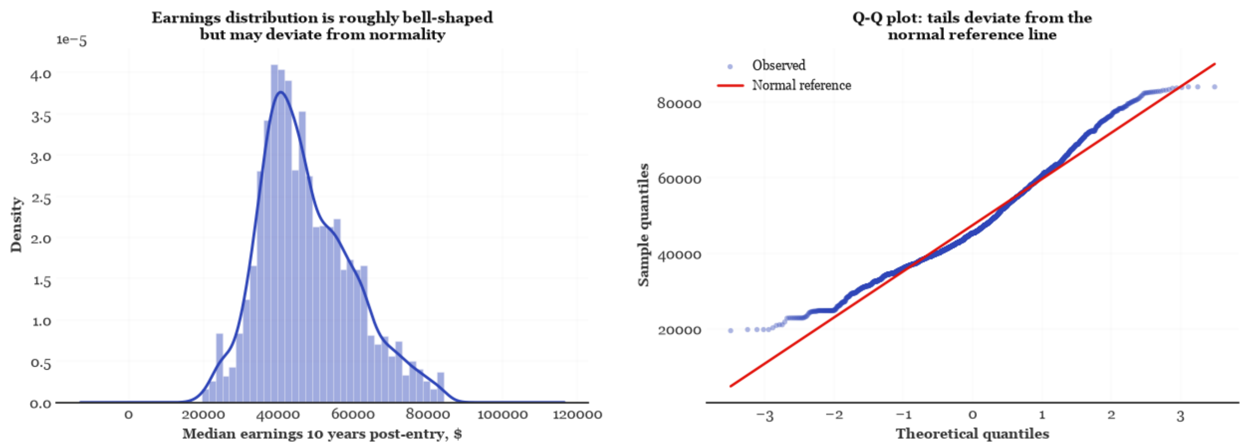

Figure 7: Left panel: histogram of median earnings with kernel density estimate. Right panel: Q-Q plot against the normal distribution. The distribution shows a mixture of three sector-level components and lighter tails than a normal curve.

Therefore, we reject the null hypothesis of zero association ($H_0 : r_s = 0$). While there is a statistically significant association between higher tuition and higher earnings, this correlation is moderate in strength and does not imply causation. As the subsequent regression models demonstrate, this aggregate rank correlation is a weighted average across distinct sector-level cost structures rather than a uniform price-to-return mechanism.

6.4 Regression Results

The three models form a nested sequence. Model 1 estimates the pooled association. Model 2 adds sector interactions. Model 3 tests whether the sector differences persist after adjusting for enrollment, degree level, and state fixed effects. Table 6 compares the fit statistics across all three models.

Table 6: Model Fit Comparison

Model	N	R^2	Adj. R^2	Log RMSE	AIC	BIC
Model 1	2,894	0.187	0.187	0.236	−153.0	−141.1
Model 2	2,894	0.398	0.397	0.203	−1,013.5	−977.7
Model 3	2,894	0.591	0.582	0.167	−2,021.8	−1,657.6

Note. Log RMSE is the root mean squared error of the log-transformed residuals. All models use HC3 standard errors.

6.4.1 Model 1: Unconditional Association

Model 1 estimates a pooled tuition elasticity of 0.121 across all 2,894 institutions, with a 95% confidence interval of [0.113, 0.129] and $p < .001$. The model explains 18.7% of the variation in log earnings. The Breusch-Pagan test rejects homoskedasticity ($p = .007$), confirming the need for HC3 standard errors.

Translating to practical terms, a student comparing a \$6,000 public tuition against a \$34,000 private nonprofit tuition would face roughly a fivefold difference. The model predicts about a 0.9% earnings advantage for the higher-tuition institution, or roughly \$400 on a \$45,000 median starting salary. The relationship is statistically reliable but economically modest. The residual diagnostics in Figure 8 show heavier tails than a normal distribution and widening residual spread at higher fitted values, confirming that the pooled model fits poorly across the range of the data.

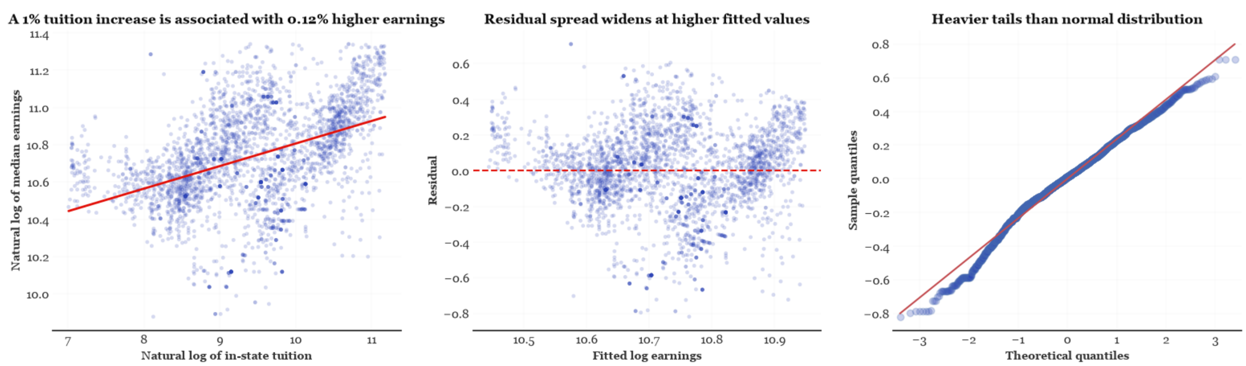

Figure 8: Model 1 diagnostics. Left panel: log-log scatter with the pooled regression line. Center panel: residuals against fitted values, showing heteroskedasticity. Right panel: Q-Q plot of residuals with heavier tails than a normal distribution.

This pooled estimate addresses the first part of the research question. A significant positive relationship exists between tuition and earnings. But the exploratory analysis has already shown that the three sectors operate on different cost-outcome curves, and the residual diagnostics confirm that a single slope cannot represent them accurately. The second part of the research question requires the sector interaction model.

6.4.2 Model 2: Sector Interaction

Adding sector indicators and tuition-by-sector interactions raises the adjusted R-squared from 0.187 to 0.397. The log RMSE drops from 0.236 to 0.203. The AIC falls from -153 to $-1{,}013$. Accounting for sector more than doubles the explained variance.

The estimated tuition elasticities are 0.194 for public institutions (95% CI [0.176, 0.213]), 0.277 for private nonprofits (95% CI [0.242, 0.312]), and 0.400 for for-profits (95% CI [0.301, 0.499]). Each slope is positive with $p < .001$. The joint Wald test for the interaction terms yields a statistic of 30.122 with $p < .001$, confirming that the tuition-earnings slope differs across sectors. This directly answers the second part of the research question: the relationship is not uniform.

The range is practically meaningful. A student comparing identical tuition levels would expect roughly twice the earnings increment at a for-profit institution compared to a public institution. Figure 9 shows the fitted lines in log-log space. The for-profit line is the steepest, but it operates in a region of the data where for-profit institutions are concentrated in particular states, offer predominantly sub-baccalaureate credentials, and enroll smaller student bodies. Their high apparent elasticity is a composition effect, not a genuine sector advantage.

6.4.3 Model 3: Institutional-Control Robustness

Model 3 adds undergraduate enrollment, predominant degree, and state fixed effects to the Model 2 specification. The adjusted R-squared rises to 0.582, and the log RMSE falls to 0.167. The added controls are jointly significant with Wald $p < 10^{-265}$, confirming that institutional characteristics beyond tuition explain a substantial share of earnings variation.

After adjustment, the tuition elasticities converge to a narrow range. Table 7 reports the estimates and confidence intervals for Models 2 and 3.

The central result is the collapse of the for-profit elasticity from 0.400 to 0.119, a reduction of roughly 70%. The private nonprofit estimate falls from 0.277 to 0.147. The public estimate remains essentially unchanged at 0.189. Figure 10 visualizes this shift.

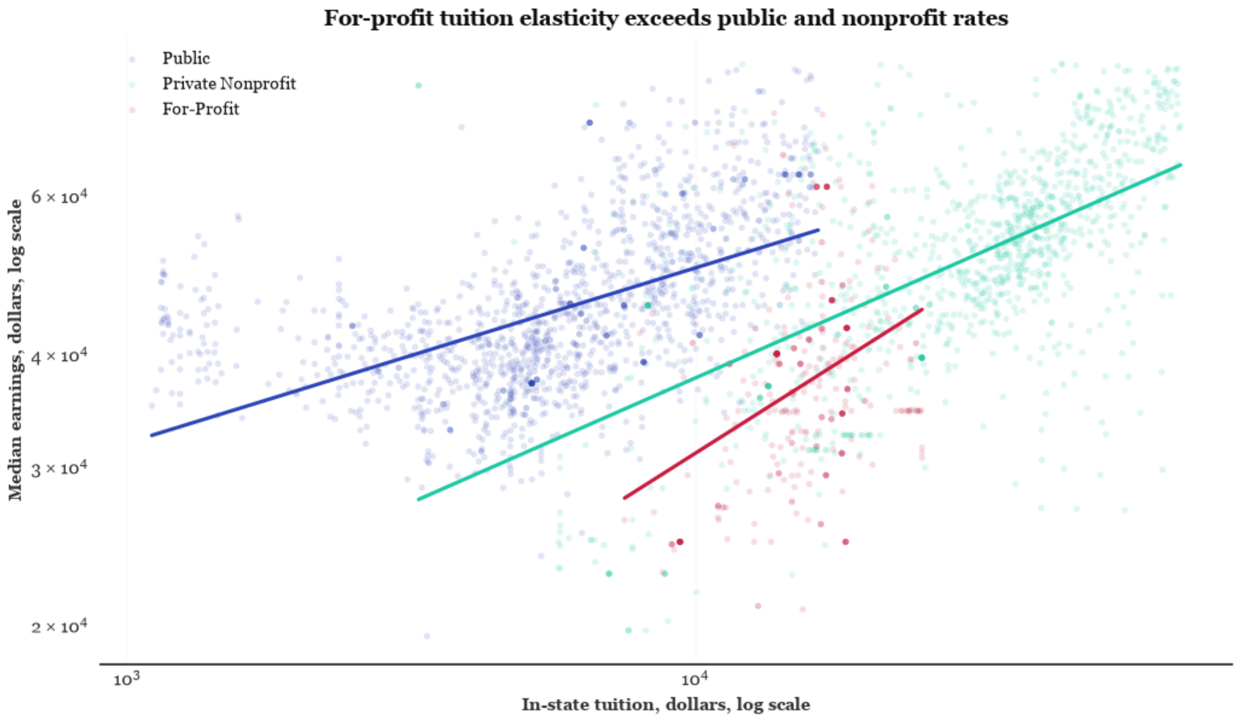

Figure 9: Log-log scatter of tuition and earnings by sector with Model 2 fitted lines. The for-profit line is the steepest but reflects composition rather than a genuine sector advantage. Public and nonprofit lines show more moderate slopes.

Table 7: Sector-Specific Tuition Elasticities with 95% Confidence Intervals

	Model 2		Model 3	
Sector	**Elasticity**	**95% CI**	**Elasticity**	**95% CI**
Public	0.194	[0.176, 0.213]	0.189	[0.168, 0.210]
Private Nonprofit	0.277	[0.242, 0.312]	0.147	[0.116, 0.178]
For-Profit	0.400	[0.301, 0.499]	0.119	[0.034, 0.205]

Note. All elasticities are from log-log OLS models with HC3 standard errors. Model 3 includes controls for log enrollment, categorical degree level with bachelor's as reference, and state fixed effects.

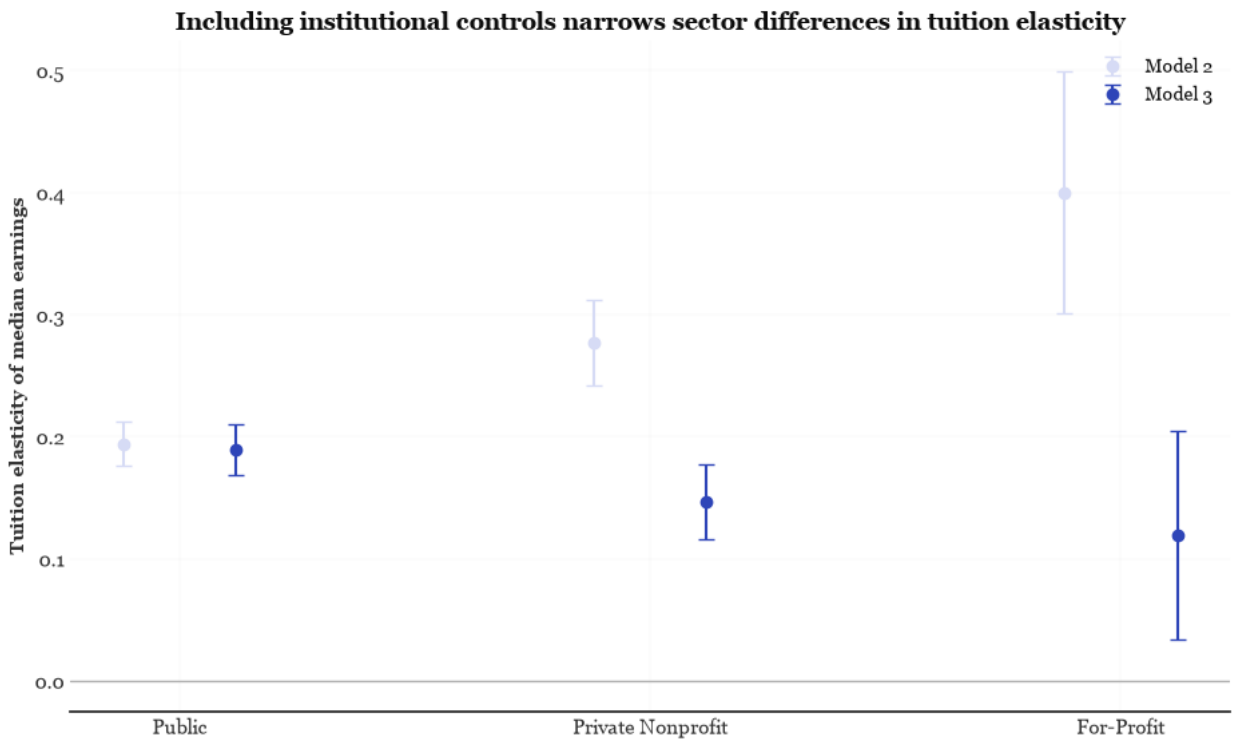

Figure 10: Tuition elasticity estimates by sector for Model 2 (unadjusted) and Model 3 (with controls). Error bars show 95% confidence intervals. The sector differences narrow substantially after adding institutional controls.

The joint tuition-by-sector interaction test yields p = 0.047 in Model 3, falling just below the conventional 0.05 threshold. The remaining sector differences, while statistically detectable, are narrow in practical terms. The three elasticities span a range of 0.07 percentage points. For a student facing a $10,000 tuition difference between two institutions, the predicted annual earnings gap is $189 at a public institution, $147 at a nonprofit, and $119 at a for-profit. Whether a range of $70 in predicted annual earnings across sectors constitutes a meaningful policy difference is a judgment the data cannot make.

The controls themselves explain roughly 19 additional percentage points of earnings variance beyond Model 2. Institutional location, size, and degree mix matter more for earnings than the tuition price. This finding reframes the research question: the answer is that the relationship exists and differs by sector, but the differences are substantially smaller than the unadjusted estimates suggest, and the practical magnitude of those differences is modest.

6.5 Sensitivity Analysis: Admission Rate

Admission rate data are available for only 1,466 of the 2,894 institutions in the analytical sample. The reporting subset overrepresents private nonprofits (908 institutions) and public institutions (503 institutions) while severely underrepresenting for-profit entities (55 institutions). This severe contraction occurs because open-enrollment for-profit institutions typically do not employ traditional admissions screening or report formal acceptance rates (Pepinsky, 2018). Two models estimated on this restricted sample test whether adding selectivity alters the core tuition-earnings parameters.

Within the reporting subset, Model 3 without admission rate produces tuition elasticities comparable to full-sample estimates for public and nonprofit institutions, whereas the for-profit elasticity shifts to -0.096 ($SE = 0.194, p = .618$). Adding linear admission rate leaves public ($\beta = 0.175, p < .001$) and private nonprofit ($\beta = 0.128, p < .001$) elasticities positive and statistically significant, while the for-profit estimate records $\beta = -0.043$ ($SE = 0.193, p = .823$). With a large standard error rendering the coefficient statistically indistinguishable from zero ($p > .05$), the negative point estimate represents a statistical artifact of severe sample underpowering rather than a meaningful inverse economic relationship. Admission rate itself records a significant negative coefficient ($\beta = -0.103$, $p < .001$), confirming that institutional selectivity is an independent predictor of post-graduation earnings.

The sector interaction terms are not jointly significant in the selectivity-adjusted model ($p = .155$). Failing to reject sector homogeneity ($p > .05$) on this restricted sub-sample suggests that once selectivity and student composition are held constant, remaining cross-sector elasticity differences are largely eliminated. However, because selective institutions are disproportionately likely to report admissions criteria, this restricted sub-sample suffers from non-random selection bias (Muse & Muse, 2024; Pepinsky, 2018). Consequently, while the sensitivity analysis confirms the stability of tuition returns among public and nonprofit colleges, restricted for-profit estimates cannot be generalized to the broader postsecondary sector.

6.6 Answering the Research Question

The research question asked two things. First, is there a significant relationship between in-state tuition and median post-graduation earnings? The answer is yes. Every model finds a positive and statistically significant association in every sector, before and after adjustment. Second, does this relationship differ across public, private nonprofit, and for-profit institutions? The answer is yes, but with an important qualification. The unadjusted sector differences are large: for-profit elasticities appear to be roughly double the public estimate in Model 2. Once enrollment, degree level, and state fixed effects are held constant, those differences narrow to a range of 0.07 elasticity points. The sector interaction remains marginally significant at p = 0.047, but the economic magnitude of the remaining gap is small.

The practical implication is that the tuition-earnings relationship is real but not a straightforward guide for decision-making. Students choosing between institutions will find more signal in the sector itself, the state where the institution is located, and the type of degree offered than in the tuition price alone. For policymakers, the narrow range of elasticities across sectors after adjustment suggests that regulatory attention to tuition pricing may be less productive than attention to the structural characteristics that predict earnings independent of price.

7 Insights and Conclusions

7.1 Insights

Tuition alone explains little

Model 1 establishes that the unconditional relationship is real but thin. The pooled tuition elasticity of 0.121 (95% CI [0.113, 0.129], $p < .001$) is statistically unambiguous across 2,894 institutions, yet it accounts for only 18.7% of the variation in log earnings. Four-fifths of the difference in what graduates earn has nothing to do with what their institution charges. The Spearman coefficient of 0.464 from the preliminary inference tells the same story in rank terms: a moderate, reliably non-zero association, not a deterministic one.

Sector is the stratifying variable, not the price

Adding institutional control and its interaction with tuition raises the adjusted R-squared from 0.187 to 0.397, and the joint HC3 Wald test on the interaction terms is decisive (statistic 30.122, $p < .001$). Sector alone contributes roughly the same explanatory weight as tuition itself. The unadjusted slopes appear to invert the intuitive ordering: for-profits post the steepest elasticity at 0.400, ahead of private nonprofits at 0.277 and public institutions at 0.194. Taken at face value, this would suggest that the sector with the weakest earnings outcomes converts tuition into earnings most efficiently.

That inversion is composition, not return

Model 3 dissolves it. Once undergraduate enrollment, predominant degree, and state fixed effects are held constant, the for-profit elasticity falls from 0.400 to 0.119, meaning roughly 70% of the unadjusted coefficient was absorbed by observable institutional structure. For-profits cluster in particular states, enroll small student bodies, and award predominantly sub-baccalaureate credentials; each of these depresses earnings independently of price, and the unadjusted interaction was reading those features as a tuition effect. After adjustment the ordering reverses to 0.189 for public, 0.147 for private nonprofit, and 0.119 for for-profit institutions. The sector difference survives, but only marginally: the joint interaction test returns $p = .0465$, and the restricted selectivity sample cannot reproduce it at all ($p = .155$). Muse and Muse (2024, pp. 35–37) document the same admission-coverage limitation for institution-level earnings regressions.

Where an institution sits matters more than what it charges

The control block adds 18.5 percentage points of adjusted explanatory power beyond Model 2, almost exactly what tuition contributed on its own, and the controls are jointly significant at $p < 10^{-265}$. For a prospective student, the practical reading is that geography and credential type carry at least as much information about likely earnings as the sticker price does. Even so, 41.8% of the variation remains unexplained (adjusted R-squared of 0.582), which is the expected ceiling for institution-level aggregates that cannot observe student background, field of study, or local labour demand. Faber (2017, pp. 45–47) identifies the same causal and omitted-variable boundaries for institution-level College Scorecard regressions.

Elasticity ranking and dollar return are not the same ranking

Elasticities are scale-free, so a larger coefficient does not mean a better deal. Priced at each sector's median institution, a 10% tuition increase costs $598 at a public institution and returns roughly $822 in predicted annual earnings, about $1,374 per $1,000 of additional tuition. The same proportional increase costs $3,393 at a private nonprofit and returns roughly $767, about $226 per $1,000, and $1,633 at a for-profit for roughly $447, about $274 per $1,000. Public institutions convert price into predicted earnings around six times more efficiently than private nonprofits, precisely because they start from a low tuition base. This is a descriptive per-dollar ratio rather than a payback calculation, since tuition recurs annually across a degree while the earnings figure is a single-year median at ten years post-entry.

The sector gap is a level gap, not a slope gap

The median private nonprofit graduate out-earns the median for-profit graduate by \$14,718. Closing that gap through tuition alone, at the for-profit sector's own adjusted elasticity of 0.119, would require the median for-profit institution to charge roughly sixteen times its current tuition, above \$260,000. The sectors sit on different intercepts, and no plausible movement along a within-sector slope crosses the distance between them. Combined with the debt-to-earnings ratios of 0.29 for public graduates against 0.45 and 0.43 for the other two sectors, the net return picture favours the public sector on both the numerator and the denominator.

7.2 Conclusion

This study sought to determine whether a significant relationship exists between published in-state tuition and post-graduation earnings, and the empirical results provide a nuanced, tempered answer. While there is a statistically significant positive association ($r_s = 0.464$, $p < .001$), the magnitude of this relationship is only modest to weak-to-moderately positive. The baseline OLS tuition elasticity of 0.121 indicates that a 100 percent increase in published tuition is associated with only a 12.1 percent increase in median post-graduation earnings, explaining less than 19 percent of baseline earnings variance ($R^2 = 0.187$).

Furthermore, this modest relationship varies across institutional sectors and weakens significantly once institutional controls are introduced. In the fully controlled model, public institutions exhibit a modest tuition elasticity of 0.189, private nonprofits record an elasticity of 0.147, and for-profit institutions display a weak elasticity of 0.119. These sub-unity elasticities confirm diminishing returns to tuition pricing across all sectors, demonstrating that higher sticker prices yield only minor percentage gains in graduate earnings.

These findings carry key practical and policy implications. For prospective students, the results demonstrate that published tuition should not be interpreted as a strong proxy for expected earnings, as institutional sector, credential level, and geographic location account for far more earnings variation than tuition alone. For policymakers, the weak tuition-earnings linkage—combined with substantial graduate debt burdens in private and for-profit sectors—underscores the need for greater transparency and cost-accountability in higher education finance.

Overall, while the null hypothesis of zero association is rejected, the empirical return on published tuition is strictly modest, heavily sector-dependent, and constrained by substantial unexplained outcome variation.

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